

Who reaches Chile’s public hospitals with private insurance? Severity and admission profile of Isapre/Particular vs FONASA discharges (GRD 2021-2024)

C1 · From the W1 project to defensible evidence — Business Intelligence (IIB423T-1), Universidad del Desarrollo
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DRAFT v0.1 · 29 September 2026

Draft status. Text in yellow boxes is information only the team can provide (W1 record, instructor feedback, contributions, AI use, reflections) or a check that still has to be run. Every number in this draft comes from the outputs saved in `01_grd_severity_analysis.ipynb`; nothing was recomputed from the raw files.

1. Summary

Chile’s open FONASA GRD files describe hospital discharges from the **public** hospital network only. Inside that network, most discharges belong to FONASA beneficiaries, but a small group of privately insured patients (*Isapre* or *Particular*) also appears (between about 1.0% and 1.9% of analyzed discharges per year). This report asks whether those privately insured patients arrive with a different **severity profile** and **admission pattern** than FONASA beneficiaries, and whether that difference is stable between 2021 and 2024.

Main observed patterns (Section 5): (i) in 2022–2024 privately insured discharges have a **lower** share of high-severity cases than FONASA discharges (about 19.3–19.5% vs 24.2–25.0%), stable across the three years; (ii) privately insured discharges are **more often admitted through the emergency department** in every year (5.5–8.7 percentage points higher); (iii) **2021 breaks the severity pattern** (28.7% vs 26.9%, i.e. private *higher*), and the group’s severity share falls by 9.4 points between 2021 and 2022, much more than FONASA’s (2.2 points). The data cannot say *why* these patients are in the public network, whether the differences reflect age or diagnosis mix, or anything about privately insured patients who never use public hospitals. The result is therefore a bounded, descriptive answer, and the report states what further evidence would be needed.

2. Problem, foundation and evolution from W1

2.1 Problem, population, period and intended use

Element	Definition used in C1
Problem	Little is known, from open data, about the case mix of privately insured patients who are hospitalized in Chile’s public network, and whether they should be monitored as a distinct group.
Service / population	Hospital discharges (<i>egresos</i>) from public hospitals coded with IR-GRD and published by FONASA as annual GRD files.
Place	Chile, public hospital network (national files; no establishment-level analysis in C1).
Period	Discharge files 2021, 2022, 2023, 2024 (four annual files).
Proposed use	Descriptive evidence to support a <i>proposed</i> decision by a public-network management analyst (see Section 3, Step 1). Not a clinical study and not a comparison of the public and private systems.

2.2 Documentary support

- **What GRD data are and how they are used in Chile.** IR-GRD (International Refined DRG) groups hospitalized *and* ambulatory episodes into base groups that are split into three severity levels and three mortality-risk levels [R5]. FONASA has used GRD as a purchasing/payment mechanism for public hospitals since 2020, with a program reform in 2023 that defined a single hospital cluster and recognized higher hospital complexity than in earlier years [R3]. IR-GRD is the system used in Chile in both the public and private sectors [R4]. These sources establish why GRD files are a plausible, standardized record of case mix, and that coverage and rules changed across the years we study.
- **Why the data cover only the public network.** The W1 decision to abandon a public-vs-private hospital comparison rests on the fact that open GRD files only include the public network. TEAM: add the FONASA institutional presentation (August 2024) cited in the notebook, with title, URL and the page/slide that supports this statement.
- **Pandemic context for 2021.** During 2020–2021 FONASA arranged a “GRD agreement” with private providers to move FONASA patients to private clinics [R6]. A regulation issued in April 2020 also made the Undersecretariat of Care Networks responsible for centrally directing patients regardless of insurance through the bed-management unit (UGCC), and set a tariff for Isapre, Capredena, Previred or private-pay patients treated in public establishments [R7]. Both are relevant to how privately insured patients may have reached public hospitals in 2021 (see Section 5.4). TEAM: confirm which La Tercera article the W1 record refers to; the one cited here is the closest match found on 29 Sep 2026.

2.3 Policy context, numerical target and measurable indicator

The National Health Strategy (*Estrategia Nacional de Salud*, ENS) 2021–2030 is the reference policy document. Two editions matter: the first version published in 2022 (four health objectives, seven strategic axes) [R2], and the updated edition formalized by Exempt Resolution No. 831 of 11 June 2024, dated October 2025 in the document, which reorganizes the strategy into nine strategic objectives, including *Equity* (OE6), *Health system* (OE8) and *Quality of care* (OE9) [R1]. The updated edition keeps as decade-level objectives improving population health, reducing health inequities and ensuring quality of care.

We keep three levels apart:

Level	Statement	Status in this project
Policy objective	Reduce inequities and assure quality of care (ENS 2021–2030).	Context only.
Numerical target	ENS targets are defined per topic (for example incidence or mortality rates).	No ENS target or indicator on severity of hospital discharges by insurance status was identified. TEAM: read ENS 2025 sections OE6, OE8 and OE9 (pp. 110–170) and confirm or correct this statement.
Indicator our data can measure	% high-severity discharges, mean IR-GRD relative weight and % emergency admissions, by insurance group and year.	Descriptive indicators of case mix, not ENS indicators and not measures of quality or equity.

This is a boundary, not a policy match: the analysis is related to the equity and health-system themes but does not measure progress toward any ENS target.

2.4 How the project evolved from W1

Stage	Question	Why it changed
W1 (original)	Do public and private hospitals treat cases of different complexity?	Rejected: open GRD data cover only the public network, so private hospitals cannot be observed.
First C1 framing	Within the public network, do FONASA beneficiaries show higher severity than privately insured ones?	Kept the available data but treated “PRIVATE” as a comparable group.
Refined C1 question	Among privately insured patients who <i>do</i> end up in the public network, is their severity profile lower, similar or higher than FONASA beneficiaries’?	Privately insured patients in these files are a self-selected subgroup (they have private coverage but were hospitalized publicly). Patients who never use the public network are unobservable, so the question is about who arrives, not about the two systems.

Instructor feedback and response. The W1 feedback had three parts. (1) For the first line (emergency-department data), the file structure had been checked but the analysis of age by type of establishment had not been done. (2) For the second line (GRD), only one record had been inspected, not the whole file. (3) The README was empty and the notebook was not fully executed, which reduced traceability. The stated priority for C1 was to turn the chosen line into a real, reproducible EDA whose results and interpretation rest directly on the data. Our response, detailed in Section 8.2:

- **Line retained: GRD.** TEAM: state in one or two sentences why the emergency-department line was not continued, e.g. no age-by-establishment analysis was available to support it.
- **Full-file inspection instead of a single record.** The C1 notebook reads the complete files of all four years (3,875,149 raw rows in total), checks column names and every line’s separator count, and profiles the full distribution of insurance category and admission type (Section 4).
- **Executed, reproducible analysis.** The notebook is run top to bottom with saved outputs, and the package includes a README (file roles, run order, versions, sources). TEAM: confirm both are true in the final archive after the fixes listed in the notebook review.

3. The seven-step map

Step	C1 answer	Where in the analysis
1. Decision	Proposed user: an analyst in a Servicio de Salud or hospital management unit. Proposed decision: whether privately insured patients in the public network deserve separate monitoring (emergency-department planning, capacity, follow-up of billing to other insurers) or can be treated as part of the general flow. Alternatives: no separate monitoring; monitor by insurance group; monitor only emergency admissions. No stakeholder was consulted; this is our proposal.	Sections 2.1, 6
2. Success criteria	Desired: knowing whether this group has a distinct, stable case-mix profile. Measurable now: severity share, relative weight and emergency share by group and year. Not measurable: cost, outcomes, quality, reason for choosing a public hospital, residence.	Section 5
3. Business questions	Main: how do severity and admission type of privately insured discharges compare with FONASA discharges in public hospitals, 2021–2024, and is the difference stable? Supporting: SQ1 gap in % high severity; SQ2 gap in mean relative weight; SQ3 gap in % emergency admissions; SQ4 does the 2021 pattern differ and why might it; SQ5 which quality or coverage differences across years limit the comparison.	Section 5
4. Business process	Insurance status, need for care, arrival (emergency, scheduled or obstetric), hospitalization in a public hospital, discharge, IR-GRD coding, annual FONASA publication. Not captured: patients who never reach a public hospital, private-sector care, outpatient and non-admitted emergency visits, why the patient came, who paid, patient residence, any bed-management referral.	Section 4.1
5. Entities, events, states	The counted event is a hospital discharge episode , not a patient, not an establishment. The same person can appear several times. Insurance group is a discharge-level attribute.	Section 4.1

6. Data representation	One row = one discharge in one annual file (129 columns, pipe-delimited). Five fields used: PREVISION, IR_29301_PESO, IR_29301_SEVERIDAD, IR_29301_MORTALIDAD, TIPO_INGRESO. No patient or establishment identifier loaded. No join (Section 4.4).	Notebook \$2-3
7. Measurement	See Table 1.	Notebook \$5

Table 1. Measures. Group = FONASA (all MAI and Libre Elección categories) or PRIVATE (ISAPRE + PARTICULAR); MILITARY (DIPRECA, CAPREDENA, CAJA DE PREVISIÓN FFAA) is kept in the prepared data but not compared.

Measure	Formula	Unit	Numerator / denominator	Exclusions	Interpretation limit
KPI 1 · % high severity	100 × discharges with IR_29301_SEVERIDAD = "3" ÷ valid discharges	% per group-year	Level 3 discharges / valid discharges of the group	Unknown insurance; rows with severity = 0 and mortality = 0; severity = DESCONOCIDO	IR-GRD has three severity levels [R5], and level 3 is taken as the highest TEAM: confirm coding in the dictionary . Severity is assigned by the grouper after discharge; it describes case mix, not risk at arrival.
KPI 2 · mean relative weight	mean of IR_29301_PESO (comma decimal converted)	weight units	Sum of weights / valid discharges	Same as KPI 1	Weights reflect expected resource use per GRD, not clinical severity.
KPI 3 · % emergency admission	100 × discharges with TIPO_INGRESO = "URGENCIA" ÷ valid discharges	% per group-year	URGENCIA / valid discharges	Same as KPI 1	"Not emergency" mixes PROGRAMADA, OBSTETRICA and DESCONOCIDO; obstetric admissions are not elective.
Gap	KPI 1 (FONASA) – KPI 1 (PRIVATE)	percentage points	—	—	Positive = FONASA higher.

4. Data, preparation and the Five Cs

4.1 Sources and what a row represents

Four annual FONASA GRD files from the course repository (2021: GRD_PUBLICO_2021.txt; 2022: GRD_PUBLICO_EXTERNO_2022.txt; 2023: GRD_PUBLICO_2023.txt; 2024: GRD_PUBLICO_2024.txt). **TEAM: add source URL, download/version date and dictionary file used.** Originals are never modified; the notebook reads only five columns. Files are pipe-delimited; 2021 and 2024 are latin1, 2022 and 2023 are UTF-16, confirmed by header inspection rather than assumed. All four have 129 columns, and the five needed columns are present in each.

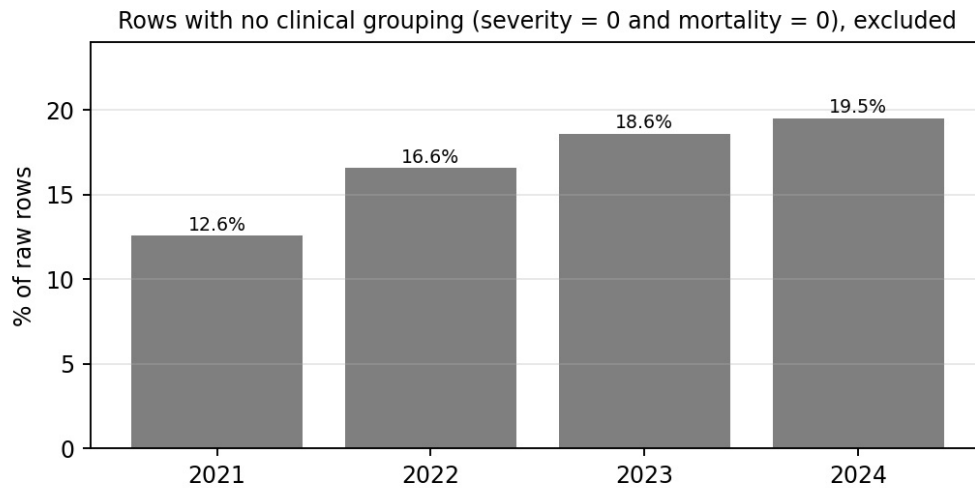
4.2 Profiling and preparation decisions

- **Malformed lines.** One malformed line in 2022 (wrong number of separators), none in the other years. It is skipped by the reader (on_bad_lines="skip"), i.e. one row out of 932,840.
- **Insurance grouping.** PREVISION values are mapped to FONASA, PRIVATE or MILITARY; DESCONOCIDO and any unmapped label become missing and are excluded.
- **Rows with no clinical grouping.** Rows with severity = 0 and mortality = 0 are excluded because severity and mortality scales are not meaningful for same-day ambulatory surgery. **TEAM: the notebook states these rows are about 100% Cirugía Mayor Ambulatoria, but no saved output shows this. Add a cell with the evidence or cite the dictionary.**
- **Emergency flag.** The valid TIPO_INGRESO categories were printed for 2024 (URGENCIA 552,102; PROGRAMADA 388,739; OBSTETRICA 144,929; DESCONOCIDO 43) and the flag was built from the observed label, not a guessed one.
- **Relative weight.** Converted from comma decimal to float; non-numeric values become missing.

Table 2. Effect of preparation on the analyzed population.

Year	Raw rows	Unmapped / unknown insurance	No clinical grouping (excluded)	Valid rows analyzed	Valid ÷ raw	FONASA	PRIVATE	MILITARY*
2021	816,909	11,601 (1.42%)	102,938 (12.6%)	703,387	86.1%	687,623	13,060	2,704
2022	932,840	59	154,774 (16.6%)	777,989	83.4%	760,887	13,841	3,261
2023	1,039,587	24	193,550 (18.6%)	845,984	81.4%	831,947	10,636	3,401
2024	1,085,813	39	212,051 (19.5%)	873,711	80.5%	861,638	8,435	3,638

*MILITARY = valid rows minus FONASA minus PRIVATE (derived). Raw minus the two exclusion columns does not equal the valid count because the filters overlap (in 2021 the valid count exceeds the subtraction by 1,017 rows) and rows with severity = DESCONOCIDO are also removed. **TEAM: add a proper sequential waterfall (raw → unmapped insurance → no grouping → severity unknown → valid).**



Rows excluded for having no clinical grouping, as a share of raw rows. The exclusion grows from 12.6% to 19.5%.

The excluded share rises every year, so the analyzed population is a shrinking fraction of the raw file, and it is defined by the same rule in all years. Whether the exclusion affects the two insurance groups equally has not been checked (see limitations).

4.3 The Five Cs

C	Result	Status
Clean	One malformed line (2022). No duplicate check on a record key was possible because no identifier was loaded. The notebook's duplicate counts (798,322 in 2021 up to 1,066,117 in 2024) come from comparing rows on only five categorical fields, so they are expected to be huge and are not evidence of duplicated discharges .	Partial. TEAM: check the 129-column dictionary for an episode identifier and rerun the duplicate check on it.
Consistent	Encodings and column names verified per year. Severity and mortality codes are the same fields in all years. Open: the relative-weight table or IR-GRD version may differ between years (mean FONASA weight is 1.19 in 2021 and 1.05–1.07 afterwards), so weights are compared within each year, not across years.	Partial. TEAM: confirm IR-GRD version per year in the dictionary or FONASA technical notes.
Conformed	The 2022 file is named "EXTERNO", unlike the others. Its insurance distribution is FONASA-dominated, like 2024. However , ISAPRE + PARTICULAR make up 1.61% of raw 2022 rows vs 0.87% in 2024, almost double, and the analyzed PRIVATE count is highest in 2022 (13,841). "Similar structure" is supported, "no difference in the private share" is not. In 2021 there are 11,601 rows with unmapped insurance versus about 50 in other years, which suggests different labels in that file.	Open. TEAM: list the unmapped 2021 PREVISION labels and decide whether they belong to PRIVATE or FONASA; check what "EXTERNO" means in FONASA's file documentation.
Current	Historical files, latest year 2024. Suitable for describing 2021–2024 patterns, not for a real-time operational decision; a monitoring use would need the annual publication schedule.	Applicable, reported.
Comprehensive	Covers coded discharges of public hospitals only. Raw rows grow 33% from 2021 to 2024 (816,909 to 1,085,813), which may reflect more hospitals or more complete coding after the 2023 program reform [R3], not only more activity. Unobserved: privately insured patients who never use public hospitals, private-sector care, patient identity and residence.	Partial. The file coverage per year is not verified.

4.4 Joins

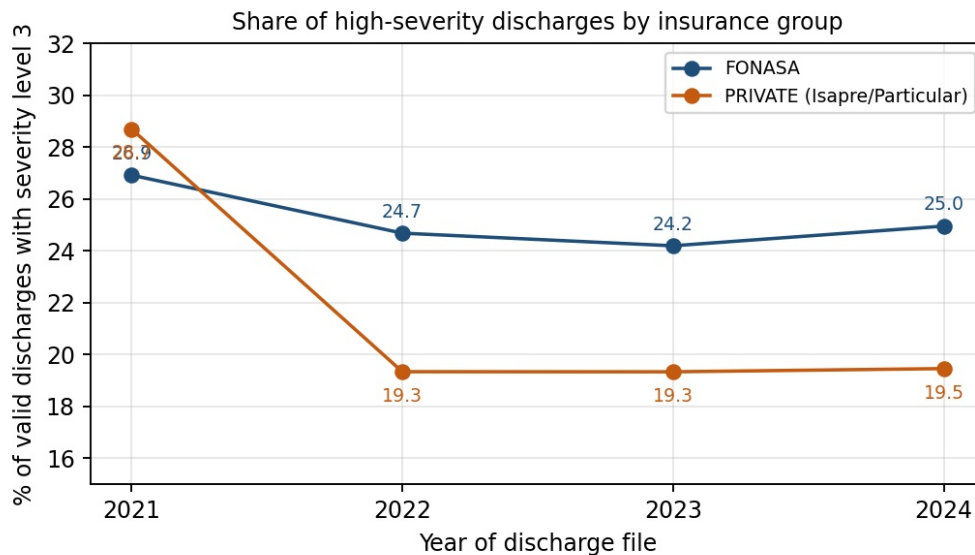
No join was performed. The question is answered with fields native to the GRD file, and the establishment table would add a location of care, which is not a patient's residence or a population denominator. A join was therefore not justified. **TEAM: state in one line whether the establishment table was actually explored in W1 and why it was dropped.**

5. Exploratory results and interpretation

Table 3. KPIs by insurance group and year (valid discharges).

Year	% high severity FONASA	% high severity PRIVATE	Gap (pp)	Mean weight FONASA	Mean weight PRIVATE	% emergency FONASA	% emergency PRIVATE
2021	26.92	28.70	–1.78	1.190	1.262	64.22	72.88
2022	24.68	19.33	5.35	1.059	0.987	59.41	64.92
2023	24.19	19.33	4.86	1.055	1.001	60.48	66.34
2024	24.95	19.45	5.50	1.068	1.045	62.32	69.28

5.1 Severity



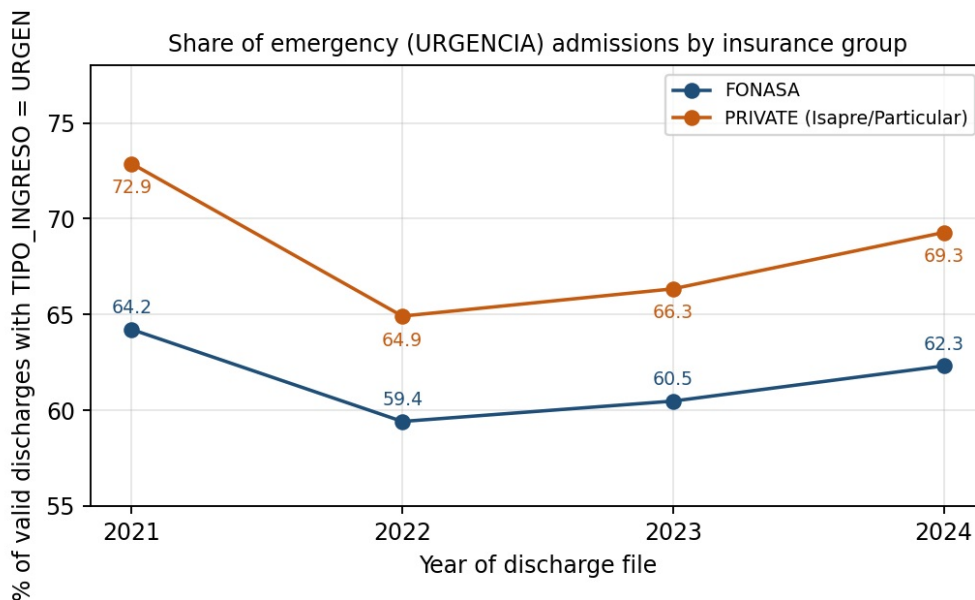
Share of valid discharges with severity level 3, by insurance group.

Observed. From 2022 to 2024 the PRIVATE group has a lower high-severity share than FONASA, with a gap of 4.9 to 5.5 points that barely moves. Relative weight tells a related but weaker story: the PRIVATE mean is 0.07, 0.05 and 0.02 below FONASA in 2022, 2023 and 2024, so the gap in weight narrows almost to parity by 2024 while the severity gap does not.

Interpretation. The pattern does not support the idea that privately insured patients reach the public network only for the most severe cases. It is compatible with several explanations that these data cannot separate: a different age or diagnosis mix, a different share of obstetric or short stays, different coding practice, or different reasons for using a public hospital. These remain hypotheses.

Comparability. Both groups are measured with the same rule and the same exclusions each year. Sampling noise is small relative to the gaps (with 8,435 PRIVATE discharges in 2024 a rough binomial margin is under 1 point), but because the file is close to a census of coded discharges, the concern is **systematic** difference, not chance.

5.2 Emergency admission



Share of valid discharges admitted through URGENCIA.

Observed. PRIVATE discharges have a higher emergency share than FONASA in all four years, by 8.7, 5.5, 5.9 and 7.0 points.

Interpretation. Combined with Section 5.1, privately insured patients in these files arrive *more often through the emergency department but with a lower share of level-3 severity* from 2022 onward. This is a descriptive contrast between two measures, and it is not evidence that the emergency use is inappropriate or that the private network failed to resolve those cases.

5.3 Group size

PRIVATE discharges are 1.9%, 1.8%, 1.3% and 1.0% of analyzed discharges in 2021–2024, and their count falls from 13,841 (2022) to 8,435 (2024) while FONASA discharges keep growing. A shrinking, small group can be more sensitive to a few hospitals or coding practices; no establishment-level check was possible without the join.

5.4 The 2021 anomaly

In 2021 the PRIVATE group has a *higher* severity share than FONASA (–1.78 gap) and a higher weight (1.26 vs 1.19). The notebook’s first explanation is that elective surgery was postponed at the peak of the pandemic, leaving proportionally more true hospitalizations: 2021 has a lower share of no-grouping rows (12.6% vs 16.6–19.5%). That would raise severity shares in **both** groups, but the change from 2021 to 2022 is –2.2 points for FONASA and –9.4 points for PRIVATE. The explanation therefore fits FONASA better than PRIVATE. A second hypothesis is that in 2020–2021 centralized bed management sent privately insured patients to public hospitals when private capacity was saturated [R7], which would select more severe private patients. Neither hypothesis can be tested with five fields. The unusually high count of unmapped insurance labels in 2021 (11,601) was noticed and has not been investigated, and it could itself affect the PRIVATE count.

The emergency measure does **not** reverse in 2021, so the anomaly is specific to severity and is not a general data defect of that year.

6. Conclusions within the evidence

What the evidence supports. In the public hospital network, between 2022 and 2024, privately insured discharges show a lower high-severity share and a higher emergency-admission share than FONASA discharges; both patterns are stable. 2021 differs in severity.

What it does not support. It does not show risk, quality of care, appropriateness of emergency use, or causal effects of insurance. Counts of discharges do not measure need. It says nothing about privately insured patients who never enter the public network, so it cannot compare the public and private systems.

Implications for the proposed decision. The two measures together suggest that privately insured patients in the public network are not simply “the sickest”: a monitoring unit would be better served by tracking both severity and admission route than by assuming either. Before any operational use the following are needed: age, sex and diagnostic group to check case mix; establishment identifiers to test whether the pattern is driven by a few hospitals; the exclusion effect by insurance group; and clarity on 2021 (unmapped labels and referral rules).

7. Project continuity beyond C1

Nothing in this section is implemented; it explains why the line can continue.

7.1 A concrete future machine-learning task

- **Task (supervised classification).** Predict whether a discharge is **high severity** (`IR_29301_SEVERIDAD = “3”`), to identify early which admissions are likely to be resource-intensive. Decision value: capacity and case-mix planning for the same proposed user as in Section 3.
- **Unit of analysis.** One discharge episode, as in this report.
- **Inputs available at the decision moment (to confirm in the dictionary).** Insurance group, `TIPO_INGRESO`, year/month, and, if present among the 129 columns, age, sex, region and establishment. **TEAM: list the actual columns available at admission, from the dictionary.**
- **Not usable as inputs (recorded only after the outcome).** GRD assignment and relative weight, mortality level, diagnosis and procedure codes, length of stay and complications. Severity itself is assigned at discharge, so a prospective model needs a clear “decision time”.
- **Additional data that may be needed.** Establishment attributes (via the establishment table), and patient-level history, which open data probably do not offer.
- **Uncertainties to review next.** Class balance (about 19–25% positives), 2021 as a structural break, hospital effects, coding-practice differences, and whether pre-discharge fields carry enough signal.
- **Exploratory alternative (no outcome labels).** Clustering hospital-year case-mix profiles (needs the establishment field).

7.2 Dashboard proposal

- **Proposed users.** A Servicio de Salud or hospital management analyst; a FONASA GRD analyst.
- **Decisions supported.** Whether to monitor non-FONASA patients separately; where emergency admissions of privately insured patients concentrate.
- **Core indicators.** Discharges (n) by insurance group; % level-3 severity; mean relative weight; % emergency admission; share excluded as “no grouping”.
- **Comparisons and filters.** FONASA vs PRIVATE (MILITARY optional); year; admission type; region and establishment if the join is added; severity level.
- **How the EDA shapes it.** Every indicator is shown with its denominator; a visible note flags 2021 and the rising excluded share; small cells are flagged, since PRIVATE has under 9,000 discharges in 2024; severity and emergency share appear side by side because they move differently.

8. Process record

8.1 Decisions and checks

Stage	Decision	Alternative considered	Reason and evidence	Check performed and result	Location
W1	Drop public-vs-private hospital comparison	Compare hospital complexity across sectors	Open GRD covers only the public network	Sources reviewed (Section 2.2)	Notebook \$4
C1	Refine the question to	Keep FONASA vs PRIVATE	Self-selection: only those	Limitation stated in	Notebook \$4

	privately insured patients <i>inside</i> the public network	as a general comparison	who reach public care are observed	Sections 2.4, 6	
C1	Verify encoding per year	Assume one encoding	Wrong encoding gives garbled column names	Header check: 129 columns, no missing columns in any year	Notebook \$2
C1	Skip malformed lines	Repair by hand	Only one line in 2022	Malformed-line count per year	Notebook \$2
C1	Keep 2022 “EXTERNO” file	Exclude 2022	Similar insurance distribution to 2024	Distribution compared; private share differs (1.61% vs 0.87%)	Notebook \$2; Section 4.3
C1	Exclude rows with severity 0 and mortality 0	Keep them	Scales not meaningful for ambulatory surgery	TEAM: add evidence cell	Notebook \$3
C1	Build emergency flag from observed labels	Hardcode “URGENCIA”	Avoid guessing a label	Category counts printed	Notebook \$3
After W1 feedback	Continue GRD line and inspect the complete files	Keep inspecting a single record; continue the emergency-department line	Instructor feedback (Section 8.2)	Full read of 3,875,149 rows over four years; per-line separator check	Notebook \$2–3
C1	No join with establishments	Join to add location	Answerable with GRD fields; location of care is not residence	Justified in Section 4.4	Notebook, final note
C1	Treat the 2021 anomaly as context, not a defect	Drop 2021	Emergency KPI does not reverse; cause unresolved	Two hypotheses stated, none tested	Notebook \$6; Section 5.4

8.2 Instructor feedback and response

Feedback received on W1. **TEAM: add date and who gave it.**

#	Advice received	Response	Reason	Change made	Evidence location
1	Line 1 (emergency data): file structure verified, but age by establishment type not analyzed.	TEAM: accepted / adapted / rejected. Line not continued in C1.	TEAM: reason	Project continues with the GRD line only.	Section 2.4
2	Line 2 (GRD): only one record inspected, not the whole file.	Accepted.	Inspecting one record cannot show encodings, malformed lines, category distributions or coverage across years.	Full files of 2021–2024 are read and profiled: header check per file, separator count on every line, insurance and admission- type distributions, exclusions per year (Table 2).	Notebook \$2–3; Section 4
3	README empty; notebook not fully executed, which reduces traceability.	Accepted.	Reviewer must be able to inspect results before deciding on a rerun.	Executed notebook with saved outputs; README with roles, order, versions and sources. TEAM: confirm in final archive.	README.md; notebook
4	Priority: a real, reproducible EDA with interpretation grounded in the data.	Accepted.	Core of C1.	KPI tables by year and group, figures, interpretation with alternative explanations and limits.	Sections 5–6

8.3 Individual contributions

Member	Concrete contribution	Evidence location
Miguel Figueroa	TEAM: actual contribution	TEAM
Sebastián Bunzli	TEAM: actual contribution	TEAM

8.4 AI use and verification

Tool	Purpose	What was accepted, modified or rejected	Verification
Claude (Anthropic)	Drafting this report from the notebook outputs and the C1 brief; searching for retrievable references	TEAM: state what you accepted, changed or rejected, and add relevant prompt/output excerpts	References were retrieved on 29 Sep 2026; TEAM: describe your own check of each claim
TEAM: any tool used for the code/analysis			

8.5 Individual reflection

Member	Contribution	Learning gain	Unresolved uncertainty
Miguel Figueroa	TEAM	TEAM	TEAM
Sebastián Bunzli	TEAM	TEAM	TEAM

References

- **[R1]** Ministerio de Salud de Chile, Subsecretaría de Salud Pública, DIPLAS. *Estrategia Nacional de Salud para el Cumplimiento de los Objetivos Sanitarios 2021–2030* (updated edition, October 2025; Res. Ex. N°831, 11 June 2024). <https://www.minsal.cl/wp-content/uploads/2026/04/Estrategia-Nacional-de-Salud->

[2021-2030.pdf](#) (accessed 29 Sep 2026).

- **[R2]** Ministerio de Salud de Chile, DENS/DIPLAS. *Estrategia Nacional de Salud para los Objetivos Sanitarios al 2030* (2022 edition; Res. Ex. N°528, 3 June 2021). <https://www.minsal.cl/wp-content/uploads/2022/03/Estrategia-Nacional-de-Salud-2022-MINSAL-V8.pdf> (accessed 29 Sep 2026).
- **[R3]** *Lecciones y desafíos del uso de los grupos relacionados por el diagnóstico en Chile*. Dialnet. <https://dialnet.unirioja.es/servlet/articulo?codigo=9734585>
TEAM: add authors, journal and year from the page.
- **[R4]** *Importancia del sistema GRD para alcanzar la eficiencia hospitalaria*. Revista Médica Clínica Las Condes, 2018. <https://www.sciencedirect.com/science/article/pii/S0716864018300592>
- **[R5]** Metamodelo SpA. *GRD más codificados de compra de servicios por parte de Fonasa*, 14–15 Feb 2023 (technical document from a GRD service provider; secondary source). <https://www.metamodelo.cl/2023/02/14/grd-mas-codificados-derivados-por-fonasa-a-clinicas-privadas/>
- **[R6]** La Tercera. *Coronavirus: 9.600 afiliados a Fonasa han sido trasladados a prestadores privados*, 17 Apr 2021. <https://www.latercera.com/la-tercera-sabado/noticia/coronavirus-9600-afiliados-a-fonasa-han-sido-trasladados-a-prestadores-privados/WKV4HA5VHVBSJL2TIUASBW2TYQ/>
- **[R7]** 24horas. *Resolución dispone precios máximos para pacientes con COVID-19 que se atiendan en clínicas*, 16 Apr 2020 (news report of the regulation; the primary text should be cited if available). <https://www.24horas.cl/coronavirus/resolucion-dispone-precios-maximos-para-pacientes-con-covid-19-que-se-atendan-en-clinicas-4103955>
- **[R8]** TEAM: FONASA institutional presentation, August 2024, cited in W1 and notebook: title, author, URL and access date.
- **[R9]** FONASA. GRD annual files 2021–2024 (course repository BI_T-main) and their data dictionary. TEAM: exact source, version and retrieval date.